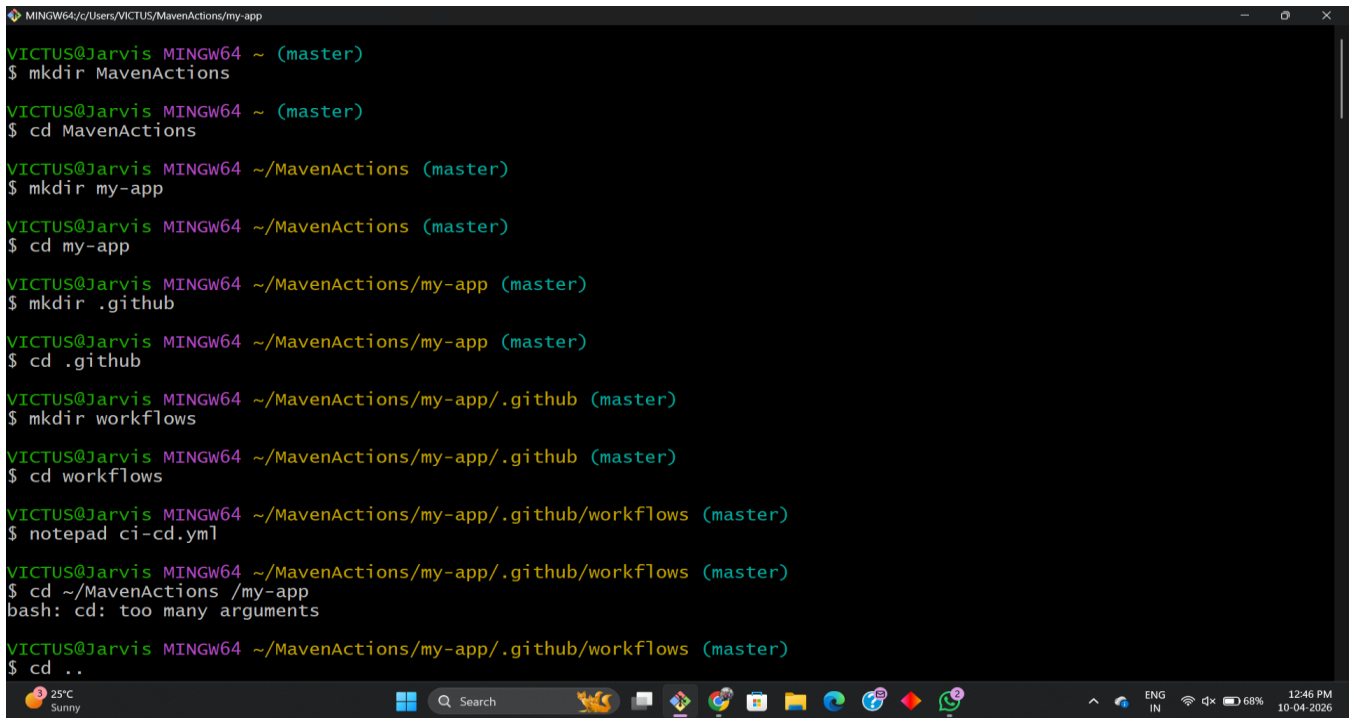


Prac-04

You are required to create a workflow configuration file:(.GitHub/workflows/ci.yml) that perform the following tasks:

1. checks out the source code from the repo
2. Sets up the appropriate java development kit(jdk) version required for the project.
3. Install all the necessary dependencies using maven
4. Compiles the application and execute unit tests using appropriate Maven command
5. Archive the generated build artifacts(such as JAR or WAR files) for the future use.

A terminal window titled 'MINGW64/c/Users/VICTUS/MavenActions/my-app' showing a series of commands and their outputs. The user creates a directory 'MavenActions', moves into it, creates 'my-app', moves into it, creates '.github', moves into it, creates 'workflows', moves into it, and creates a file 'ci-cd.yml' using 'notepad'. The terminal shows the current directory path at each step. The Windows taskbar is visible at the bottom with the date 10-04-2026 and time 12:46 PM.

```
MINGW64/c/Users/VICTUS/MavenActions/my-app
VICTUS@Jarvis MINGW64 ~ (master)
$ mkdir MavenActions

VICTUS@Jarvis MINGW64 ~ (master)
$ cd MavenActions

VICTUS@Jarvis MINGW64 ~/MavenActions (master)
$ mkdir my-app

VICTUS@Jarvis MINGW64 ~/MavenActions (master)
$ cd my-app

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ mkdir .github

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ cd .github

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github (master)
$ mkdir workflows

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github (master)
$ cd workflows

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github/workflows (master)
$ notepad ci-cd.yml

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github/workflows (master)
$ cd ~/MavenActions /my-app
bash: cd: too many arguments

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github/workflows (master)
$ cd ..
```

STEP-01: Create the folder and builds the template

STEP-02: Get inside the folder

STEP-03: Make the directory .github/workflows

STEP-04: Create the file named ci-cd.yml with the content in it.

```
MINGW64/c/Users/VICTUS/MavenActions/my-app
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github/workflows (master)
$ cd ..

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app/.github (master)
$ cd ..

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git init
Initialized empty Git repository in C:/Users/VICTUS/MavenActions/my-app/.git/

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git add .

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ ls

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ ls -a
./ ../ .git/ .github/

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ mvn archetype:generate -DgroupId=com.example -DartifactId=my-webapp0 -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 35 kB/s)
```

STEP-05: Git init the folder

git add .

Git commit -m "message"

Git remote origin <url of git hub repo>

Git push origin main

```
MINGW64/c/Users/VICTUS/MavenActions/my-app
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ mvn archetype:generate -DgroupId=com.example -DartifactId=my-app -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] -----
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: packageInPathFormat, Value: com/example
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
```

STEP-06: Visit the repo go to the Actions Tab and see the workflow is running.

```
MINGW64/c/Users/VICTUS/MavenActions/my-app
[INFO] Parameter: artifactId, Value: my-app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Archetype in dir: C:\Users\VICTUS\MavenActions\my-app\my-app
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.169 s
[INFO] Finished at: 2026-04-10T12:35:37+05:30
[INFO] -----

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ ls
my-app/ my-webapp0/

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ ren my-webapp0
bash: ren: command not found

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ rm -r my-webapp0

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ ls
my-app/

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git init
Reinitialized existing Git repository in C:/Users/VICTUS/MavenActions/my-app/.git/

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git add .
warning: in the working copy of 'my-app/src/main/webapp/WEB-INF/web.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my-app/src/main/webapp/index.jsp', LF will be replaced by CRLF the next time Git touches it

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git commit -m "file workflow updated"
[master (root-commit) 8502656] file workflow updated
4 files changed, 119 insertions(+)
create mode 100644 .github/workflows/ci-cd.yml
create mode 100644 my-app/pom.xml
create mode 100644 my-app/src/main/webapp/WEB-INF/web.xml
create mode 100644 my-app/src/main/webapp/index.jsp

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git remote add origin
usage: git remote add [<options>] <name> <url>

    -f, --[no-]fetch          fetch the remote branches
    --[no-]tags              import all tags and associated objects when fetching
                             or do not fetch any tag at all (--no-tags)
    -t, --[no-]track <branch> branch(es) to track
    -m, --[no-]master <branch> master branch
    --[no-]mirror[=(push|fetch)] set up remote as a mirror to push to or fetch from

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git remote add origin https://github.com/Kavya4619MGupta/my-app.git

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (master)
$ git branch -M main

VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
$ git push -u origin main
```

```
MINGW64/c/Users/VICTUS/MavenActions/my-app
$ git push -u origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (13/13), 1.90 KiB | 388.00 KiB/s, done.
Total 13 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Kavya4619MGupta/my-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ ls -a
bash: ls-a: command not found
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ ls -a
./ ../ .git/ .github/ my-app/
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ cd ~/MavenActions/my-app
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ mv my-app/* .
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ rmdir my-app
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ ls
pom.xml  src/
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ ls
pom.xml  src/
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ git add .
git commit -m "Fixed project structure"
git push
warning: in the working copy of 'src/main/webapp/WEB-INF/web.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/webapp/index.jsp', LF will be replaced by CRLF the next time Git touches it
[main 12ceb7d] Fixed project structure
 3 files changed, 0 insertions(+), 0 deletions(-)
 rename my-app/pom.xml => pom.xml (100%)
 rename {my-app/src => src}/main/webapp/WEB-INF/web.xml (100%)
 rename {my-app/src => src}/main/webapp/index.jsp (100%)
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 1.40 KiB | 719.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Kavya4619MGupta/my-app.git
 8502656..12ceb7d  main -> main
```

```
VICTUS@Jarvis MINGW64 ~/MavenActions/my-app (main)
```

```
$ |
```